HE2003 Econometrics I:
Linear Regression with One Regressor 4

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Linear Regression with One Regressor

- Suppose you evaluate an estimator many times over repeated randomly drawn samples. For one sample, the result may be larger than the true value; for another sample, it may be smaller than the true value. But it is reasonable to hope that, on average, you would get the right answer. Thus, a desirable property of an estimator is that the mean (center) of its distribution equals μ; if so, the estimator is said to be unbiased.
- Suppose you have two candidate estimators, $\tilde{Y}$ and $\hat{Y}$, both of which are unbiased. How might you choose between them? One way to do so is to choose the estimator with the tightest distribution. That is, we should pick the estimator with the smallest variance. If $\tilde{Y}$ has a smaller variance than $\hat{Y}$, then $\tilde{Y}$ is said to be more efficient than $\hat{Y}$.
- The sample average $\bar{Y}$ is an unbiased estimator of μ: $E[\bar{Y}] = \mu$. Next, suppose we want to compare Y_1 and $\bar{Y}$. Notice that Y_1 is an unbiased estimator with $\mathrm{Var}[Y_1] = \sigma^2$. On the other hand, $\mathrm{Var}[\bar{Y}] = \sigma^2/n$. Therefore, $\bar{Y}$ is more efficient than Y_1 for $n \geq 2$.

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Concerning the efficiency of the sample average $\bar{Y}$, we have the following important result.

KEY CONCEPT: $\bar{Y}$ is best linear unbiased estimator (BLUE)

Let $\hat{Y}$ be any linear unbiased estimator of the population mean μ. That is, $\hat{Y} = \sum_{i=1}^{n} a_i Y_i$, where $a_1, ..., a_n$ are constants (e.g., if we take $a_1 = 1$, and $a_2 = 0, ..., a_n = 0$, then $\hat{Y} = Y_1$), and $E[\hat{Y}] = \mu$. Then,

$$\mathrm{Var}[\bar{Y}] \leq \mathrm{Var}[\hat{Y}].$$

- Note that $\hat{Y}$ is a linear combination of $Y_1, ..., Y_n$.
- The BLUE result states that if $\hat{Y}$ is a linear unbiased estimator of the population mean μ, then the variance of $\hat{Y}$ cannot be smaller than that of the sample average $\bar{Y}$.
- That is to say, $\bar{Y}$ is the most efficient estimator of μ among all possible constructions of linear unbiased estimators.

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Example: The efficiency of $\bar{Y}$.

- We have seen that Y_1, the first observation of data, is also an unbiased estimator of the population mean μ.
- However, in general it is less efficient than $\bar{Y}$ because $Var(Y_1) = \sigma^2$ while $Var(\bar{Y}) = \sigma^2/n$. If the sample size $n = 10$, then $Var(Y_1)$ is 10 times as large as $Var(\bar{Y})$.
- To emphasize this point, the following Table contains the outcome of a computer simulation study: 20 random samples of size 10 ($n = 10$) were repeatedly generated from a normal distribution, with mean equal to 2 and variance equal to 1 ($\mu = 2$ and $\sigma^2 = 1$).
- For each of the 20 random samples, we compute the two estimators of μ: Y_1 and $\bar{Y} = (Y_1 + Y_2 + ... + Y_{10})/10$; These values are listed in the following table. e.g., the first row corresponds to Y_1 and $\bar{Y}$ computed from the first random sample, the second row computed from the second random sample, etc.

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We note that: (i) the values for Y_1 are much more spread out than those for $\bar{Y}$: Y_1 ranges from -0.64 to 4.27, while $\bar{Y}$ from 1.16 to 2.58; (ii) in 16 out of 20 cases, $\bar{Y}$ is closer than Y_1 to the true value $\mu = 2$.

Replication	y_1	$\bar{y}$
1	−0.64	1.98
2	1.06	1.43
3	4.27	1.65
4	1.03	1.88
5	3.16	2.34
6	2.77	2.58
7	1.68	1.58
8	2.98	2.23
9	2.25	1.96
10	2.04	2.11
11	0.95	2.15
12	1.36	1.93
13	2.62	2.02
14	2.97	2.10
15	1.93	2.18
16	1.14	2.10
17	2.08	1.94
18	1.52	2.21
19	1.33	1.16
20	1.21	1.75

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- The average of Y_1 across the 20 simulations is about 1.89, while that for $\bar{Y}$ is 1.96. These averages are close to 2 because both Y_1 and $\bar{Y}$ are unbiased estimators of μ ($= 2$ in this example).
- But in terms of efficiency, the sample average $\bar{Y}$ is far superior to Y_1 as an estimator of μ, and such superiority will increase with n: as the variance of $\bar{Y}$ is equal to σ^2/n, it will decrease with n; by contrast, the variance of Y_1 is always σ^2.

Linear Regression with One Regressor

Now, we are able to state and discuss the important Gauss-Markov Theorem.

- This theorem justifies the use of the OLS method rather than using a variety of other estimators.
- OLS is unbiased: With more involved algebra (than for the case with sample average), we can show that

  $$E[\hat{\beta}_0] = \beta_0, \;\; E[\hat{\beta}_1] = \beta_1.$$

  However, there exist many other unbiased estimators of β_0 and β_1.
- Might there be other linear unbiased estimators with variances smaller (more efficient) than the OLS estimators?

Under the OLS assumptions, the answer is No. The Gauss-Markov Theorem shows that the efficiency result of $\bar{Y}$ can be extended to the OLS estimators: The OLS estimators are the most efficient among all linear unbiased estimators of β_0 and β_1.

Linear Regression with One Regressor

KEY CONCEPT (Gauss-Markov Theorem): The OLS estimators are best linear unbiased estimators (BLUE)

Under the OLS assumptions (and some other technical assumptions), the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the best linear unbiased estimators of β_0 and β_1, respectively.

- The Gauss-Markov Theorem states that among all the linear unbiased estimators of β_0 and β_1, the OLS estimator has the smallest variance (the most efficient).
- That is, for any alternative estimators $\tilde{\beta}_0$ and $\tilde{\beta}_1$ that are linear and unbiased, we have

$$Var[\hat{\beta}_0] \leq Var[\tilde{\beta}_0], \;\; Var[\hat{\beta}_1] \leq Var[\tilde{\beta}_1].$$

- Therefore, when the OLS assumptions hold, we need not look for alternative linear unbiased estimator: None will be better than OLS.

Linear Regression with One Regressor

Empirical Application: The Capital Asset Pricing Model (CAPM) and the "Beta" of a Stock

- A fundamental idea of modern finance is that an investor needs a financial incentive to take a risk. Therefore, the expected return on a risky asset, R, should be higher than that on a safe/risk-free asset, R_f. So the expected excess return, $R - R_f$, should be positive.
- According to the CAPM, the expected excess return $R - R_f$ is proportional to the expected excess return on a portfolio of all assets (the market portfolio) $R_m - R_f$:

$$R - R_f = \beta(R_m - R_f) \tag{1}$$

- In practice, R_f is often taken to be the rate of interest on short-term US government bond. According to CAPM, a stock with $\beta < 1$ ($\beta > 1$) has less (more) risk than the market portfolio and also a lower (higher) expected excess return.

Linear Regression with One Regressor

- The "beta" of a stock has become a workhorse of the investment industry, and we can obtain estimated betas for hundreds of stocks on investment firm websites.
- Those betas typically are estimated by OLS regression of the actual excess return on the stock (Y) against the actual excess return on a broad market index (X) such as the S&P 500.
- The table below gives estimated betas for seven US stocks.

Company	Estimated β
Wal-Mart (discount retailer)	0.1
Coca-Cola (soft drinks)	0.6
Verizon (telecommunications)	0.7
Google (information technology)	1.0
General Electric (industrial)	1.1
Boeing (aircraft)	1.3
Bank of America (bank)	1.7

Linear Regression with One Regressor

KEY CONCEPT: Regression with a Binary Variable

The discussion for observational data so far has mainly focused on the case that the regressor is continuous (e.g., the STR). Regression can also be used when the regressor is binary - it takes only two values, 0 or 1. e.g., X might be a worker's gender (=1 if female, =0 if male), whether a school district is urban or rural (=1 if urban, =0 if rural), or whether the class size is small or large (=1 if small, =0 if large).

Let us first consider the following model with a binary regressor:

$$wage_i = \beta_0 + \beta_1 D_i + u_i, i = 1, ..., n$$

where $wage_i$ is the wage of i-th person and D_i is a binary regressor (dummy variable/indicator variable) that satisfies

$$D_i = \begin{cases} 1 & \text{if } i\text{-th person is female;} \\ 0 & \text{if } i\text{-th person is male.} \end{cases}$$

Linear Regression with One Regressor

- Then, β_1 is the coefficient for investigating if there is gender gap in wage.
- In particular, $E[u_i|D_i] = 0$ by OLS Assumption 1, and we have:

$$\begin{aligned} E[wage_i|D_i = 1] &= \beta_0 + \beta_1, \\ E[wage_i|D_i = 0] &= \beta_0. \end{aligned}$$

- Notice that $E[wage_i|D_i = 1]$ is the average wage of the group of women, and $E[wage_i|D_i = 0]$ the average wage of the group of men. Thus, β_1 is the difference in the average wages between women and men:

$$\beta_1 = E[wage_i|D_i = 1] - E[wage_i|D_i = 0].$$

Linear Regression with One Regressor

Application to Test Scores

- Suppose we define a binary regressor D_i that equals either 0 or 1, depending on whether the STR is less than 20:

$$D_i = \begin{cases} 1 & \text{if the STR in district } i \text{ is less than 20;} \\ 0 & \text{if the STR in district } i \text{ is no less than 20.} \end{cases}$$

- An OLS regression of the test score against the binary regressor D using the California data set yields:

$$\widehat{\text{Test Score}} = 650.0 + 7.4 \times D.$$

- Thus, the average score for the group of districts with STR greater than or equal to 20 ($D = 0$) is 650.0, the average score for the group of districts with STR less than 20 ($D = 1$) is

$$650.0 + 7.4 = 657.4,$$

and the difference between the two groups is 7.4, the coefficient of the binary regressor D.

Linear Regression with One Regressor

- Besides the above example, binary regressors (dummy variables) have many other applications.
- For example, let us consider a model with a common slope and two different intercepts:

$$Y_i = \beta_0 + \delta_0 D_i + \beta_1 X_i + u_i, \ i = 1, ..., n, \tag{2}$$

  where D_i is a dummy variable and X_i is a continuous variable, so that $Y_i = \beta_0 + \beta_1 X_i + u_i$ when $D_i = 0$, and $Y_i = (\beta_0 + \delta_0) + \beta_1 X_i + u_i$ when $D_i = 1$.

  For the CA dataset, we may define $D_i = 1$ if the i-th school district is in Region A and $D_i = 0$ if it is in Region B.

- In fact, it is possible to include more dummy variables to further enrich our model.

Linear Regression with One Regressor

Empirical Application: Effects of Law School Rankings and GPA on Starting Salaries

The dataset *lawsch_ranking.dta* contains the average starting salaries of students in each US law school (Harvard, Yale, Columbia, etc.), ranking of these law schools (No.1-142), average college GPA of the students in each school, etc.

Now we consider the following model:

$$salary_i = \beta_0 + \beta_1 top10_i + \beta_2 r11\_25_i + \beta_3 r26\_50_i + \beta_4 GPA_i + u_i,$$

i=1, ..., n. Define the dummy variables $top10$, $r11\_25$, $r26\_50$ to be equal to 1 when the ranking of the i-th school falls into the appropriate range, and 0 otherwise.

Q: How many categories/levels in this model? Which schools belong to the base group (the group with intercept equal to β_0) in this model?

We will give further details on the estimation of this model in latter classes.

Linear Regression with One Regressor

Testing Hypothesis about the Regression Coefficients

As a starting point of statistical hypotheses testing, we specify a null hypothesis. For instance, we want to test whether the following statement is true: "On average, college graduates earn $20/hour." Let Y be the hourly earning of college graduates, then the null hypothesis is: $\mu = 20$, where μ denotes the population mean $E(Y)$.

- In general, we may consider the null hypothesis that the population mean is a specific value μ_0:

$$H_0 : \mu = \mu_0.$$

- The alternative hypothesis specifies what would happen if the null hypothesis is not true. The most general alternative hypothesis is

$$H_1 : \mu \neq \mu_0.$$

This is called a (two-sided) alternative hypothesis because it allows μ to be either less than or greater than μ_0.

Linear Regression with One Regressor

- We decide whether to reject the null hypothesis on the basis of the observations/data $Y_1, \ldots, Y_n$.
- For instance, because $\bar{Y}$ is a good estimator of μ (unbiased and efficient), if $H_0 : \mu = \mu_0$ is true, then $\bar{Y}$ will typically be close to μ_0. In contrast, if H_0 is wrong, then the difference between $\bar{Y}$ and μ_0 should be large.
- Hence, we may reject the null hypothesis ($H_0 : \mu = \mu_0$) if the difference between $\bar{Y}$ and μ_0 is larger than a prespecified value c (critical value). The question is: How do we determine c?

KEY CONCEPT: Type I and Type II Errors.

We determine c by considering the errors of the test. There are two kinds of errors in hypothesis testing: rejecting H_0 when H_0 is true (type I error) and accepting H_0 when H_1 is true (type II error).

	Accept H_0	Reject H_0
When H_0 is true	Correct Decision	Type I error
When H_1 is true	Type II error	Correct Decision

Linear Regression with One Regressor

- Review: What are the type I and II errors in a Covid-19 test?
- Typically, there is a trade-off between the type I and type II errors. We cannot minimize the probabilities of type I and type II errors simultaneously: A small type I error implies a large type II error, and vice versa.
- The standard approach for testing sets the probability of type I error to be equal to a prespecified level. The prespecified level of the type I error is called the significance level of the test. In economics, the significance level is typically chosen to be 10%, 5%, or 1% [In legal (e.g., H_0: the defendent is not guilty) or medicine settings (e.g., H_0: the new drug does not have effect), it may be better to use 0.1%].
- For a test with 5% significance level, we specify the critical value c so that $P_{H_0}(\text{Reject } H_0) = 0.05$. That is, we choose the value of c so that when H_0 is true, the probability of rejecting the null hypothesis H_0 (commiting type I error) is 5%. Here, $P_{H_0}(\text{Reject } H_0)$ refers to the probability of rejecting H_0 when H_0 is actually true.

Linear Regression with One Regressor

Application to the Test Score Data

Now, we want to test hypotheses about the slope β_1.

- Your client, the superintendent, calls you with a problem.
- She has an angry taxpayer in her office who asserts that cutting class size will not help boost test scores, so hiring more teachers is a waste of money. **Class size, the taxpayer claims, has no effect on test scores.**
- The taxpayer's claim can be restated in the language of econometrics: the causal effect on test scores of a change in class size is zero, that is $\beta_1 = 0$.
- You already provided the superintendent with an OLS estimate of β_1 using your sample of 420 observations on California school districts ($\hat{\beta}_1 = -2.28$). Given this result, can you reject the taxpayer's hypothesis that $\beta_1 = 0$, or should you accept it, at least tentatively pending further new evidence? That is, **how (statistically) significant is your estimation result?**

Linear Regression with One Regressor

Again, the linear regression model is written as:

$$Y_i = \beta_0 + \beta_1 X_i + u_i.$$

The null hypothesis and the (two-sided) alternative hypothesis about the slope parameter can be written as:

$$H_0 : \beta_1 = \beta_{1,0} \quad \text{vs.} \quad H_1 : \beta_1 \neq \beta_{1,0}.$$

- Here, $\beta_{1,0}$ is the value to be specificed by us. e.g., in the example of angry taxpayer, we want to test $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$ (we want to test if the slope parameter equals zero or not). Then, $\beta_{1,0}$ equals 0 in this example.
- We need a statistical procedure to choose between H_0 and H_1 (**to decide the taxpayer's claim can be rejected or not**). This is done by the t-test.

Linear Regression with One Regressor

- When the sample size is large (assuming the OLS assumptions are satisfied), $\hat{\beta}_1 \sim N\left(\beta_1, \sigma^2_{\hat{\beta}_1}\right)$, that is, $\hat{\beta}_1$ follows a normal distribution with mean equal β_1 and variance equal $\sigma^2_{\hat{\beta}_1}$.

  This implies that:

  $$\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0,1)$$

  That is, the standardized version of $\hat{\beta}_1$ follows $N(0,1)$, a standard normal distribution, in large samples.

- The variance of $\hat{\beta}_1$, $\sigma^2_{\hat{\beta}_1}$, can be estimated by replacing the population variance with its sample analogue:

  $$\hat{\sigma}^2_{\hat{\beta}_1} = \frac{1}{n} \frac{\frac{1}{n-2}\sum_{i=1}^{n}(X_i - \bar{X})^2 \hat{u}_i^2}{[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2]^2}, \tag{3}$$

  where $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ (**Need Not** recite the formula of $\hat{\sigma}^2_{\hat{\beta}_1}$).

Linear Regression with One Regressor

- The standard error of $\hat{\beta}_1$ is the square root of eq.(3):

$$SE(\hat{\beta}_1) = \sqrt{\hat{\sigma}^2_{\hat{\beta}_1}} = \hat{\sigma}_{\hat{\beta}_1}.$$

  It measures the spread/dispersion of the distribution of $\hat{\beta}_1$.

- Although its formula is complicated, in practical applications the standard error can be computed by regression software.
- In large samples, $SE(\hat{\beta}_1)$ will be very close to $\sigma_{\hat{\beta}_1}$, so the following result holds:

$$\frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1)$$

  in large samples.